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Final Report

Date: 30.09.2016

Patient name: **Ahmad, Amani Mustafa / Almohammed, Abdulhamide Mohammed**

Your ref.: **064230 / R6508/16**

DOB (dd.mm.yyyy): **04.07.1992 / 01.01.1981**

Sample collection date (dd.mm.yyyy): **03.09.2016**

Order received (dd.mm.yyyy): **06.09.2016**

Sex: **female / male**

Patient no.: **1150124 / 1150130**

Sample type: **blood, EDTA**

Order no.: **62322058 / 62322067**

Whole Exome Sequencing (CentoXome PLATINUM®) was performed for both probands of the couple.

Clinical information: both probands are asymptomatic; they are the parents of two deceased girls with skeletal dysplasia (most likely thanatophoric dysplasia) who died aged 2 days. A diagnosis of Desbuquois dysplasia is suspected. According to the pedigree provided, the parents are consanguineous; negative family history.

Variants with possible relevance to patient's phenotype:

Gene (transcript)	Nucleotide (protein)	Zygosity		Described by	In silico parameters*	MAF**	Variant classification***	Disorder (OMIM#, inheritance)
		Ahmad, Amani Mustafa	Almohammed, Abdulhamide Mohammed					
CANT1 (NM_001159772.1)	c.902_906dup (p.Ser303Alafs*21)	Het.	Het.	Huber, 2009 (PMID: 19853239)	Frame-shift, premature stop codon	Not described	Pathogenic (class 1)	Desbuquois dysplasia type 1 (251450, AR)

*: number of in silico prediction programs that predict pathogenicity/all applicable programs (SIFT, PolyPhen2, AlignGV, MutationTaster). **highest minor allele frequency (MAF) of representative population (Exome Aggregation Consortium database (ExAC), Exome Sequencing Project (ESP), or 1000Genome project (1000G)). *** based on ACMG recommendations.

A diagnosis of Desbuquois dysplasia type 1 for the deceased children is likely.

Given the results we recommend retrospective clinical analysis to evaluate compatibility of phenotype with the identified variant as well as genetic counselling.

Interpretation

By whole exome sequencing we **detected a heterozygous variant in the CANT1 gene, c.902_906dup (p.Ser303Alafs*21) in both probands. This variant has been confirmed by Sanger sequencing. It has been previously described as disease-causing for Desbuquois dysplasia by Huber et al., 2009 (HGMD Professional 2016.2 - PMID: 19853239). In CentoMD® 3.0 it is listed in 1 affected patients in a homozygous state and in a heterozygous state in the patient's asymptomatic parents. It is classified as pathogenic (class 1) according to the recommendations of Centogene and ACMG (please, see additional information below).**

Pathogenic variants in the CANT1 gene are causative for autosomal recessive Desbuquois dysplasia type 1

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(DBQD). DBQD is an autosomal recessive chondrodysplasia belonging to the multiple dislocation group and characterized by severe prenatal and postnatal growth retardation (stature less than -5 SD), joint laxity, short extremities, and progressive scoliosis. The main radiologic features are short long bones with metaphyseal splay, a 'Swedish key' appearance of the proximal femur (exaggerated trochanter), and advanced carpal and tarsal bone age with a delta phalanx (OMIM: 251450).

We conclude that **both probands are heterozygous carriers of a heterozygous pathogenic variant in the CANT1 gene and that therefore a diagnosis of Desbuquois dysplasia type 1 for the deceased children is likely.** We recommend a clinical re-evaluation to determine the compatibility of the detected variant with the described symptoms. Genetic counselling is recommended.

Incidental findings

We did not detect any class 1 or 2 variants in the genes for which incidental findings are reported based on the ACMG guidelines.

If you have any further questions or analysis requests, please do not hesitate to contact us at support@centogene.com.

Best regards,



Prof. Arndt Rolfs, MD
Chief Medical Director



Karen Wessel, PhD
Clinical Scientist

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Analysis statistics

	Average % target bp covered						
	coverage (X)	0X	≥ 1X	≥ 5X	≥ 10X	≥ 20X	≥ 50X
Ahmad, Amani Mustafa	157.93	0.254751	99.7452	98.9969	97.8484	95.1728	85.0906
Almohammed, Abdulhamide Mohammed	169.916	0.134534	99.8655	99.3181	98.3245	95.9405	86.7372

Centogene Mendelian variant classification
(established gene disease association)
Class 1 – Pathogenic**Class 2** – Likely pathogenic**Class 3** – Variant of uncertain significance (VUS)**Class 4** – Likely benign**Class 5** – Benign**Class 6** – Disease-associated variant**Methods**

Approximately 37 Mb (214,405 exons) of the Consensus Coding Sequences (CCS) were enriched from fragmented genomic DNA by >340,000 probes designed against the human genome (Nextera Rapid Capture Exome, Illumina) and the generated library sequenced on an Illumina NextSeq or HiSeq 4000 platform (Illumina) to an average coverage depth 100-130X. An end to end inhouse bioinformatics pipelines including base calling, primary filtering of low quality reads and probable artefacts, and annotation of variants was applied.

All disease causing variants reported in HGMD®, in ClinVar or in CentoMD® (class 1) as well as all variants with minor allele frequency (MAF) of less than 1% in ExAC database are considered. Evaluation is focused on exons and intron boundaries +/-20.

All relevant inheritance patterns are considered, provided family history and clinical information are used to evaluate eventually identified variants. Only variants related to the phenotype are reported. Any relevant variant identified by NGS are in-house validated or Sanger sequenced in forward and reverse direction to exclude NGS artefacts.

Sanger sequencing

Exon 4 of the CANT1 gene was analysed by PCR and sequencing of both DNA strands of the entire coding region and the highly conserved exon-intron splice junctions. The reference sequence is: CANT1: NM_138793.3.

Limitations

Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations in genes potentially related to the proband's medical condition are reported. Rare polymorphisms may lead to false negative or positive results. Misinterpretation of results may occur if the information provided is inaccurate or incomplete. If results obtained do not match the clinical findings, additional testing should be considered.

NGS cannot precisely read through repeat expansion regions and thus cannot yield data i.e. for repeat expansion in the Huntington gene, the SCA-genes, the myotonic dystrophy repeat region, and other similar regions.

Additional information

This test was developed and its performance validated by Centogene AG. The US Food and Drug Administration (FDA) has determined that clearance or approval of this method is not necessary and thus neither have been obtained. This test has been developed for clinical purposes. All test results are reviewed, interpreted and reported by our scientific and medical experts.

In line with ACMG recommendations for reporting of incidental findings in clinical exome and genome sequencing (Genetics in Medicine, 2013), we report incidental findings, i.e. pathogenic variants (class 1) and likely pathogenic variants (class 2) only in the recommended genes.

To also exclude mistaken identity in your clinic, several guidelines recommend testing a second sample that is independently obtained from the proband. Please note that any further analysis will result in additional costs.

The classification of variants can change over the time. Please feel free to contact Centogene (testing@centogene.com) in the future to determine if there have been any changes in classification of any reported variants.